Machine Learning in Applications
AM01 Project Report

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***Abstract*—Industrial robots generate large volumes of multivariate sensor data during operation, including joint positions, velocities, motor currents and power-related measurements. Subtle changes in these signals may indicate early degradation, such as slower movements, reduced precision or abnormal energy consumption. However, detecting such changes is challenging because anomalies are rare, often weakly separated from normal behavior and may involve only a subset of sensor channels. This project investigates reconstruction-based anomaly detection for KUKA industrial robot data, with the main objective of evaluating whether adding an adversarial component to a standard Autoencoder improves detection performance. The work further enhances the proposed architecture by implementing score fusion strategies, directly exploiting prior distribution constraints imposed by the Adversarial Autoencoder (AAE) on the latent space. Experimental results validate the efficacy of the proposed approach, achieving an overall performance improvement, successfully overcoming inherent data imbalances to demonstrate robust detection capabilities.**

I. Introduction

Industrial robots are increasingly used in manufacturing environments where repeatability, speed and precision are essential for both productivity and product quality. Modern robotic systems continuously generate multivariate sensor measurements, including joint positions, velocities, motor currents and power-related signals. These data streams contain valuable information about the operating state of the robot and can potentially reveal early signs of degradation, such as slower movements, reduced positioning accuracy, abnormal current absorption or unusual power consumption. Detecting such deviations before they lead to failures is a central objective of predictive maintenance and industrial monitoring.

Anomaly detection in this setting is challenging for several reasons. First, robot sensor data are high-dimensional and strongly correlated, since the physical movement of the robot couples multiple joints and electrical quantities. Second, anomalous behavior is rare compared with normal operation, leading to a severe class imbalance. Third, degradation may be subtle: anomalous samples may not necessarily lie outside the global range of normal values, but may instead differ through small distributional shifts or localized changes affecting only a few sensor channels. For these reasons, simple thresholding on individual raw variables is often insufficient.

This project investigates reconstruction-based anomaly detection for KUKA industrial robot data, with the main objective of comparing a standard Autoencoder (AE) with an Adversarial Autoencoder (AAE). The AE baseline follows a simple reconstruction-based approach: the model is trained to reconstruct normal robot behavior, and anomalies are identified through high reconstruction error. The AAE extends this idea by adding adversarial latent-space regularization and by combining multiple sources of anomaly evidence, including reconstruction error and latent-space abnormality. In both cases, the neural model is not used directly as a conventional binary classifier; instead, anomaly detection is performed through a score-based decision rule.

The comparison is motivated by the hypothesis that adversarial regularization may improve anomaly detection by encouraging normal samples to occupy a smoother and more constrained latent space. If this latent structure is meaningful, anomalous samples may be detected not only because they reconstruct poorly, but also because their latent representations are unlikely under the learned normal latent distribution or deviate from the imposed prior. This makes the AAE potentially more expressive than a reconstruction-only detector, although the benefit must be evaluated experimentally.

Since anomaly detection is strongly affected by class imbalance, the evaluation does not rely only on accuracy. Instead, the analysis includes point-wise precision, recall, F_1, F_β, balanced accuracy, precision-recall area under the curve and average precision. In addition, feature-selection experiments are considered to study whether removing redundant or weakly informative channels improves detection performance and interpretability. This is especially relevant for robot sensor data, where preserving the identity of the original physical variables can help explain which channels contribute most to anomalous behavior.

The remainder of this report is organized as follows. Section II introduces the background concepts related to anomaly detection, autoencoders, adversarial autoencoders, latent-space outlier detection, score calibration and evaluation under class imbalance. Section III describes the dataset, preprocessing strategy, model architectures, anomaly-score definitions and evaluation protocol. Section IV presents the experimental results, including the AE–AAE comparison, feature-selection analysis and score-distribution diagnostics. Moreover, Section V discusses the findings, methodological limitations and possible future extensions.

II. Background

Anomaly detection aims to identify observations or patterns that deviate from the expected behavior of a system. In industrial monitoring, anomalies may correspond to faults, degradation, abnormal operating conditions or early indicators of future failure. Compared with standard supervised classification, anomaly detection is often more difficult because anomalous examples are rare, heterogeneous and not always available during training. For this reason, many practical approaches are unsupervised or semi-supervised: the model is trained mainly, or exclusively, on normal data and later detects deviations from the learned normal behavior [1].

In multivariate sensor data, each observation can be represented as a vector

$$x \in R^d,$$

where each component corresponds to a sensor channel or derived physical quantity. In such data, variables are often correlated because they describe different aspects of the same physical process. An anomaly may therefore appear not only as an extreme value in one channel, but also as an unusual combination of otherwise plausible values. This motivates the use of multivariate models able to learn relationships across the complete input vector.

A. Reconstruction-based anomaly detection

A common approach to unsupervised anomaly detection is reconstruction-based modeling. The central assumption is that a model trained on normal data will learn to reconstruct normal samples accurately, while abnormal samples will be reconstructed less accurately because they do not follow the learned normal patterns. Autoencoders are widely used for this purpose because they learn a compressed latent representation of the input and then reconstruct the original signal from that representation [2].

A standard Autoencoder is composed of an encoder and a decoder. The encoder maps the input vector x to a lower-dimensional latent representation:

$$z = E_\theta(x),$$

where θ denotes the encoder parameters. The decoder maps this latent representation back to the input space:

$$\hat{x} = D_\phi(z) = D_\phi(E_\theta(x)),$$

where ϕ denotes the decoder parameters. The model is usually trained by minimizing a reconstruction loss, such as the mean squared error:

$$\mathcal{L}_{rec} = \frac{1}{N}\sum_{i=1}^{N} \|x_i - \hat{x}_i\|_2^2.$$

After training, the reconstruction error can be used as an anomaly score. In the simplest case, the score is the mean squared reconstruction error over all features:

$$s_{rec}(x) = \frac{1}{d}\sum_{j=1}^{d}(x_j - \hat{x}_j)^2.$$

A larger value of $s_{rec}(x)$ indicates that the sample is reconstructed poorly and is therefore more likely to be anomalous. Reconstruction-based scores are particularly useful when the model has been trained primarily on normal data, because the learned representation is expected to describe normal operating patterns better than abnormal ones.

B. Adversarial Autoencoders

An Adversarial Autoencoder extends the classical autoencoder by adding an adversarial regularization mechanism in the latent space [3]. In addition to reconstructing the input, the encoder is encouraged to produce latent codes whose distribution matches a chosen prior distribution, commonly a standard Gaussian:

$$p(z) = \mathcal{N}(0, I).$$

This is achieved by introducing a discriminator that tries to distinguish latent vectors sampled from the prior distribution from latent vectors produced by the encoder. The encoder is then trained adversarially to fool the discriminator. As a result, the training objective contains two complementary components: a reconstruction objective and an adversarial objective. The reconstruction component preserves the ability to represent the input data, while the adversarial component regularizes the global structure of the latent space.

In anomaly detection, this regularization may be useful because it can force normal samples to occupy a smoother and more constrained latent distribution. In principle, anomalous samples may then produce less consistent latent representations, poorer reconstructions or latent codes that are unlikely under the learned normal latent distribution. However, this improvement is not guaranteed. The adversarial component can also make optimization more difficult and introduces additional hyperparameters, such as discriminator capacity and learning rates. Therefore, the effect of the adversarial component must be evaluated empirically.

C. Latent-space outlier detection

Autoencoders and adversarial autoencoders provide not only reconstructions but also latent representations. These latent vectors can be analyzed directly for anomaly detection. If the encoder maps normal samples into a compact or high-density region of latent space, then latent vectors lying far from this region may indicate abnormal behavior.

One way to formalize this idea is density-based outlier detection. Let $z = E_\theta(x)$ be the latent representation of a sample. A density estimator can be fitted to latent vectors obtained from normal training samples. Samples assigned low density by this estimator are then considered more anomalous. Conceptually, this gives an outlier score of the form

$$s_{out}(x) = -\log \hat{p}(z),$$

where $\hat{p}(z)$ is an estimated latent-space density. Larger values of $s_{out}(x)$ correspond to lower estimated density and therefore stronger evidence of abnormality.

Related approaches include kernel density estimation, Gaussian mixture models, robust covariance estimators and local density or neighborhood-based methods such as Local Outlier Factor [4]. These methods differ in how they model the structure of normal latent vectors, but they share the same general principle: anomalies are expected to lie in sparse, low-probability or locally inconsistent regions of the representation space.

Kernel Density Estimation (KDE) is a non-parametric technique that estimates a continuous probability distribution by centering a smooth mathematical function, or kernel, on each observed data point and summing them together. This aggregation causes kernels in tightly clustered regions to overlap and accumulate into high-density peaks, representing high

probability. Conversely, sparse regions exhibit minimal overlap and yield low probability values, inherently facilitating the identification of outliers [5].

For adversarial autoencoders, the encoder is explicitly encouraged to match a prior distribution. If the prior is standard Gaussian, it is reasonable to choose a Gaussian Kernel.

For the same reason, an additional latent-space criterion is available. Latent vectors with large squared norm are unlikely under the prior. This motivates a prior-deviation score:

$$s_{lat}(x) = \|z\|_2^2.$$

This score is proportional to the negative log-density of a standard Gaussian up to an additive constant. It therefore measures how strongly the encoded sample deviates from the intended latent prior.

D. Score calibration and fusion

Many anomaly detectors produce continuous scores rather than direct binary labels. A binary decision requires either thresholding a single score or combining multiple scores into a calibrated decision function. Given a score $s(x)$ and threshold τ, the binary prediction is

$$\hat{y}(x) = 1[s(x) > \tau],$$

where $\hat{y} = 1$ denotes an anomaly and $\hat{y} = 0$ denotes a normal sample.

The choice of τ determines the trade-off between false positives and false negatives. A low threshold usually increases recall but may also increase the number of false alarms, while a high threshold usually improves precision but may miss subtle anomalies. Thresholds can therefore be selected in several ways, for example by using a fixed quantile of normal validation scores, by optimizing a validation metric, or by selecting a desired operating point on a validation curve.

In this work, threshold selection is based on the precision-recall curve computed on the validation set. For each candidate threshold τ, the corresponding precision and recall are computed. The selected threshold is the one whose precision-recall has the maximum F_1 score:

$$\tau = \arg\max_{\tau} F_1(Precision(\tau), Recall(\tau))$$

The final prediction rule is therefore

$$\hat{y}(x) = 1[s(x) > \tau]$$

This criterion selects a threshold that balances precision and recall on the validation data. It is particularly appropriate in imbalanced anomaly-detection settings because it calibrates the detector using the trade-off between detecting anomalies and avoiding false alarms, rather than relying on accuracy or on the score distribution of normal samples alone.

When multiple anomaly scores are available, they may capture complementary information. For example, reconstruction error measures how well a sample can be reproduced in input space, while latent density measures whether the encoded representation lies in a normal region of latent space. A score-fusion model can combine these signals into a single decision score:

$$s_{fusion}(x) = f(s_1(x), s_2(x), \ldots, s_m(x)).$$

A simple and interpretable choice is logistic regression, which learns a linear combination of the input scores and maps it to a binary decision boundary. When trained with class weighting, logistic regression can partially compensate for class imbalance by assigning greater importance to the minority class during calibration.

The threshold associated with a logistic-regression decision function is naturally zero:

$$\hat{y}(x) = 1[s_{fusion}(x) > 0].$$

Thus, score fusion replaces manual threshold selection with a supervised calibration step based on validation data.

E. Evaluation under class imbalance

Anomaly detection problems are typically highly imbalanced because normal samples are much more frequent than anomalous ones. In such settings, accuracy alone can be misleading. For example, a detector that predicts every sample as normal may achieve high accuracy if anomalies are rare, while having zero ability to detect the events of interest. For this reason, evaluation must include metrics that separately account for false positives and false negatives.

The main point-wise quantities are true positives (TP), false positives (FP), true negatives (TN) and false negatives (FN). Precision measures the reliability of anomaly alarms:

$$Precision = \frac{TP}{TP + FP}$$

Recall measures the fraction of actual anomalies that are detected:

$$Recall = \frac{TP}{TP + FN}$$

The F_1 score combines precision and recall symmetrically:

$$F_1 = 2\frac{Precision \cdot Recall}{Precision + Recall}$$

More generally, the F_β score allows recall and precision to be weighted differently:

$$F_\beta = (1 + \beta^2)\frac{Precision \cdot Recall}{\beta^2 Precision + Recall}$$

For $\beta > 1$, recall is weighted more strongly, meaning that missed anomalies are penalized more heavily than false alarms.

Balanced accuracy is also useful under class imbalance because it averages sensitivity and specificity:

$$BalancedAccuracy = \frac{1}{2}\left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP}\right).$$

In addition to threshold-dependent metrics, it is important to evaluate the quality of the continuous anomaly scores.

Precision-recall analysis is particularly relevant for imbalanced datasets. The precision-recall curve evaluates precision and recall over all possible thresholds, while the area under this curve and average precision summarize how well the score ranks anomalous samples above normal ones [6].

F. Feature selection for anomaly detection

Feature selection can improve anomaly detection when some variables are nearly constant, redundant or weakly related to the anomaly mechanism. Removing such features can reduce noise, simplify the model and improve interpretability. Unlike dimensionality reduction methods such as Principal Component Analysis, feature selection preserves the original sensor variables. This is important in industrial monitoring because selected features can still be interpreted physically, for example as joint positions, velocities, currents or power-related quantities.

Several families of feature-selection methods are relevant for multivariate anomaly detection. A first simple approach is near-constant feature removal. Features whose variance or range is close to zero contain little information for distinguishing operating conditions, and may introduce numerical instability or unnecessary model parameters. Such features can be removed by applying a small threshold to the empirical variance or range:

$$\mathrm{Var}(x_j) < \epsilon.$$

This is an unsupervised filter because it does not use class labels.

A second approach is correlation-based redundancy removal. Sensor datasets often contain variables that are strongly correlated because they describe related physical quantities or coupled system dynamics. If two features have very high absolute correlation, retaining both may add little new information while increasing the dimensionality of the input. Correlation pruning removes one feature from highly correlated pairs:

$$|\rho(x_j, x_k)| > \rho_{max},$$

where $\rho(x_j, x_k)$ is the Pearson correlation coefficient between features j and k, and ρ_{max} is a chosen threshold. This method is also unsupervised, since it uses only relationships among input features.

A third approach is distributional feature ranking. Instead of measuring only variance or redundancy, this type of method evaluates how differently each feature is distributed between normal and anomalous samples. One suitable statistic is the two-sample Kolmogorov–Smirnov statistic, defined for feature j as

$$D_j = \sup_x \left| F_{0,j}(x) - F_{1,j}(x) \right|,$$

where $F_{0,j}$ and $F_{1,j}$ are the empirical cumulative distribution functions of feature j for normal and anomalous samples, respectively. A larger value of D_j indicates stronger separation between the two empirical distributions. Ranking features by this statistic can identify variables whose marginal distributions change substantially under anomalous behavior.

A fourth approach is model-based feature ranking using reconstruction errors. Reconstruction-based anomaly detectors compute feature-wise errors:

$$e_j(x) = (\hat{x}_j - x_j)^2.$$

If a feature is consistently reconstructed much worse for anomalous samples than for normal samples, it may be informative for anomaly detection. A possible reconstruction-separation score is

$$R_j = \frac{\mu_{1,j}^{rec} - \mu_{0,j}^{rec}}{\sigma_{0,j}^{rec} + \epsilon},$$

where $\mu_{1,j}^{rec}$ is the mean reconstruction error of anomalous samples for feature j, $\mu_{0,j}^{rec}$ and $\sigma_{0,j}^{rec}$ are the mean and standard deviation of reconstruction errors for normal samples, and ϵ is a small constant for numerical stability. Features with larger R_j contribute more strongly to reconstruction-error separation between normal and anomalous observations.

These feature-selection techniques differ in the type of information they use. Near-constant filtering and correlation pruning are unsupervised and can be fitted using only normal training data. Distributional ranking and reconstruction-error ranking require labeled validation data because they compare normal and anomalous behavior. Consequently, the latter methods should be applied only after the train-validation-test split, and final performance should be reported on a separate test set. Together, these methods provide complementary views: some remove uninformative or redundant variables, while others identify features most associated with anomalous behavior.

III. Materials and methods

This section describes the methodological components used for anomaly detection: the problem formulation, preprocessing strategy, model architectures, anomaly-score definitions, calibration procedures and evaluation metrics.

A. Problem formulation

The task is formulated as point-wise binary anomaly detection on multivariate robot sensor observations. Each sample is represented as a feature vector

$$x_i \in R^d,$$

where each component corresponds to a robot sensor channel or derived state variable. The associated binary label is

$$y_i \in \{0, 1\},$$

where $y_i = 0$ denotes normal robot behavior and $y_i = 1$ denotes slow or anomalous behavior.

The models are trained in a semi-supervised anomaly-detection setting. Training uses only normal samples, while validation and test data contain both normal and anomalous samples. This reflects the practical situation in which normal operating data are easier to obtain than representative examples of all possible faults or degradation patterns.

The detector is score-based. A model first assigns each sample a continuous anomaly score $s(x)$, where larger values

indicate stronger evidence of abnormality. A binary prediction is then obtained by applying a decision rule:

$$\hat{y}(x) = 1[s(x) > \tau],$$

where τ is a threshold or decision boundary.

B. Dataset Split

To ensure a robust evaluation, the KUKA dataset is partitioned into specific subsets based on the modeling approach. For the baseline Autoencoder (AE), the data is split into training, validation, and test sets. The training set constitutes the largest partition and comprises solely normal samples, aligning with the semi-supervised learning paradigm. Conversely, both the validation and test sets contain a mixture of normal and anomaly samples.

For the Adversarial Autoencoder combined with score fusion (AAE + Fusion), the validation procedure is modified to accommodate the additional calibration steps. The standard validation data is further split into two equal-sized sets. The first set is explicitly utilized to train the logistic-regression fusion model, while the second set is reserved for hyperparameter tuning and final model selection.

C. Preprocessing

Preprocessing is fitted only on the training data and then applied to validation and test data. This prevents information from validation or test samples from influencing feature scaling or feature filtering.

The preprocessing pipeline includes feature-wise min-max normalization and optional feature filtering. Highly correlated features may be removed by computing the absolute Pearson correlation between feature pairs and discarding redundant variables above a chosen correlation threshold. Optional near-constant feature removal and percentile-based clipping are also considered in feature-selection experiments.

For min-max normalization, each retained feature j is transformed as

$$x'_{ij} = \frac{x_{ij} - m_j}{M_j - m_j},$$

where m_j and M_j are the minimum and maximum values of feature j estimated from the training data after any preceding preprocessing steps. Extremely small feature ranges are replaced by a constant value to avoid numerical instability.

D. Autoencoder detector

The baseline detector is a fully connected Autoencoder. The encoder maps the input vector x to a latent representation z, and the decoder maps z back to the input space:

$$z = E_\theta(x),$$

$$\hat{x} = D_\phi(z) = D_\phi(E_\theta(x)).$$

The encoder has the following structure:

$$d \to 72 \to 64 \to 56 \to 48 \to 32 \to 24 \to 16 \to \ell,$$

where d is the input dimension after preprocessing and ℓ is the latent dimension. The decoder mirrors this structure:

$$\ell \to 16 \to 24 \to 32 \to 48 \to 56 \to 64 \to 72 \to d.$$

Each hidden layer consists of a linear transformation followed by a ReLU activation, while the latent and output layers are linear.

The Autoencoder is trained by minimizing the mean squared reconstruction loss:

$$\mathcal{L}_{rec} = \frac{1}{B} \sum_{i=1}^{B} \|x_i - \hat{x}_i\|_2^2,$$

where B is the batch size.

For anomaly detection, feature-wise squared reconstruction errors are computed:

$$e_j(x) = (\hat{x}_j - x_j)^2, \quad j = 1, \ldots, d.$$

The errors are sorted in descending order:

$$e_{(1)} \geq e_{(2)} \geq \cdots \geq e_{(d)}.$$

The AE anomaly score is the mean of the largest k feature errors:

$$s_{AE}(x) = \frac{1}{k} \sum_{r=1}^{k} e_{(r)}.$$

This top-k score emphasizes localized reconstruction failures, which is useful when abnormal behavior affects only a subset of sensor channels.

The AE decision threshold τ_{AE} is calibrated using validation samples. This is chosen to yield the (recall, precision) pair closest to the ideal top-right corner of the PR curve.

The AE prediction rule is therefore

$$\hat{y}_{AE}(x) = 1[s_{AE}(x) > \tau_{AE}].$$

E. Adversarial Autoencoder detector

The Adversarial Autoencoder uses the same encoder and decoder structure as the AE, but adds a discriminator in the latent space. The discriminator receives latent vectors and outputs a scalar logit. It is trained to distinguish latent vectors sampled from a standard Gaussian prior from latent vectors produced by the encoder:

$$z_{real} \sim \mathcal{N}(0, I),$$

$$z_{fake} = E_\theta(x).$$

The AAE training procedure alternates between reconstruction and adversarial optimization. First, the encoder and decoder are updated using the reconstruction loss. Second, the discriminator is updated to separate prior samples from encoded samples. Third, the encoder is updated so that encoded latent vectors become more difficult for the discriminator to distinguish from prior samples. This encourages the latent representation of normal data to follow the chosen prior distribution.

The adversarial component uses a hinge-style objective. For discriminator output $D(z)$, the discriminator terms are:

$$\mathcal{L}_{D,real} = \max(0, 1 - D(z_{real})),$$

$$\mathcal{L}_{D,fake} = \max(0, 1 + D(z_{fake})).$$

The discriminator is additionally regularized with an R1 gradient penalty:

$$\mathcal{L}_{R1} = \frac{\gamma}{2} E_{z_{real}} \left[\|\nabla_{z_{real}} D(z_{real})\|_2^2 \right].$$

The encoder adversarial objective is:

$$\mathcal{L}_G = \max(0, 1 - D(z_{fake})).$$

After training, the AAE detector combines three sources of anomaly evidence.

The first is the top-k reconstruction score:

$$s_{rec}(x) = \frac{1}{k} \sum_{r=1}^{k} e_{(r)}.$$

The second is a latent-space outlier score. A density estimator is fitted on latent representations of normal training samples. For a new sample with latent vector $z = E_\theta(x)$, the outlier score is:

$$s_{out}(x) = -\log \hat{p}(z),$$

where $\hat{p}(z)$ is the estimated latent-space density. Larger values indicate lower density and therefore stronger evidence of abnormality.

The third is a prior-deviation score:

$$s_{lat}(x) = \|z\|_2^2.$$

Since the adversarial prior is a standard Gaussian, this score measures how far the encoded sample lies from the center of the intended latent distribution.

The three scores are combined into a score vector:

$$u(x) = \begin{bmatrix} s_{rec}(x) \\ s_{out}(x) \\ s_{lat}(x) \end{bmatrix}.$$

A logistic-regression fusion model is trained on the split of validation data to combine these components into a single signed anomaly score:

$$s_{AAE}(x) = w^\top u(x) + b.$$

Since the logistic regression model only consists of 4 parameters (i.e. the weight vector and the bias), regularization is not needed thus leaving with no hyperparameters to optimize.

The AAE decision boundary is:

$$\hat{y}_{AAE}(x) = 1[s_{AAE}(x) > 0].$$

Thus, the AE uses a reconstruction-only score with quantile calibration, while the AAE uses a fused score combining reconstruction and latent-space information.

TABLE I: Comparison of the AE Baseline against the proposed AAE + Fusion model.

Model	Precision	Recall	F1-Score	PR AUC
AE Baseline	0.688	0.794	0.737	0.850
AAE + Fusion	0.932	0.956	0.943	0.976

TABLE II: Ablation study comparing the standalone AAE, AE + Fusion, and AAE + Fusion models.

Model	Precision	Recall	F1-Score	PR AUC
AAE (Standalone)	0.873	0.863	0.868	0.932
AE + Fusion	0.883	0.867	0.875	0.948
AAE + Fusion	0.932	0.956	0.943	0.976

IV. Results and Discussion

This section presents the experimental results, evaluating the performance of the proposed Adversarial Autoencoder (AAE) combined with a score fusion strategy. The evaluation uses multivariate sensor data from KUKA industrial robots and focuses on metrics like precision, recall, the F_1-score, and the Precision-Recall Area Under the Curve (PR AUC) to handle the severe class imbalance typical in anomaly detection.

A. AE Baseline vs. AAE + Fusion

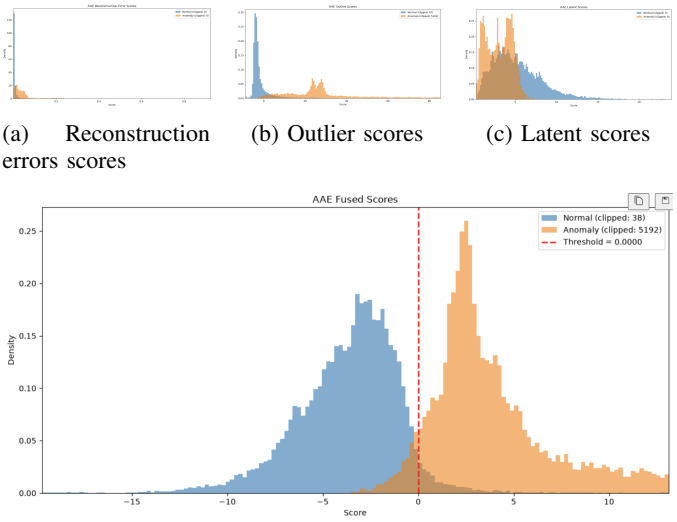
The primary objective of this project was to evaluate how well an Adversarial Autoencoder (AAE) performs compared to a standard Autoencoder (AE) baseline. For this comparison, the AE baseline was built using the top 40 features selected by the reconstruction separation statistic, which provided the most effective starting point for the standard model.

As shown in Table I, the AAE + Fusion model achieves much higher scores across all evaluated metrics. This shows that adding adversarial regularization and combining different anomaly scores, specifically reconstruction error, latent-space outlier density, and prior deviation, creates a much stronger detector than using reconstruction alone.

B. Ablation Study

An ablation study was performed to measure the specific contributions of the adversarial regularization and the score fusion mechanism. The baseline feature selection was already shown to be helpful in earlier exploratory phases. Table II details the performance of the standalone AAE, the AE with score fusion, and the combined AAE + Fusion model to evaluate these individual parts.

The results in Table II highlight the value added by each design choice. The comparison between the standalone AAE and the AAE + Fusion approach shows that the score fusion mechanism greatly increases the recall rate, which improves the overall F_1-score. The contribution of each score can be visualized in Fig 1. Furthermore, comparision between the fused AE model against the fused AAE model proves that the AAE + Fusion model performs better across all metrics. This confirms the idea that adversarial regularization helps normal samples form a smoother, more constrained latent space (Figure 2), making anomalies much easier to detect.



(a) Reconstruction errors scores (b) Outlier scores (c) Latent scores

Fig. 1: Scores obtained from the individual sources (top row) and their fusion into the final combined scores (bottom row).

TABLE III: Effect of varying the top-k parameter on the F_1-score.

Top-k	F1-Score
$k = 1$	0.944
$k = 2$	0.944
$k = 3$	0.942
$k = 5$	0.940
$k = 9$	0.937

Additionally, we investigated the AAE + Score Fusion model's sensitivity to the k parameter within the top-k reconstruction error formulation (Table III). Experimental results indicated that varying k yields only marginal changes in overall performance, with F_1-scores remaining remarkably stable. Notably, the specific case of $k = 9$ corresponds to the total number of features retained after the initial near-constant removal preprocessing step. Thus, this value represents the mean reconstruction error across the entire available feature set rather than a restricted top-k subset. This demonstrates that while isolating the largest feature errors is conceptually beneficial, the proposed architecture is highly robust and does not require exhaustive fine-tuning of k to maintain high detection accuracy.

V. Conclusions and Future Works

This project studied a reconstruction-based anomaly detection method for high-dimensional, imbalanced sensor data from KUKA industrial robots. The main goal was to see if an adversarial component and score fusion strategies could improve the performance of a standard Autoencoder.

The experimental results clearly support the proposed approach. The baseline AE, which used the top 40 features selected by the reconstrucion separation-statistic, provided a good starting point but struggled to find subtle changes in the data distribution. On the other hand, the AAE + Fusion model performed much better, reaching an F_1-score of 0.943 and a PR AUC of 0.976. The ablation study proved that both the adversarial regularization and the combination of multiple anomaly scores are needed to successfully detect early degradation while keeping false alarms low.

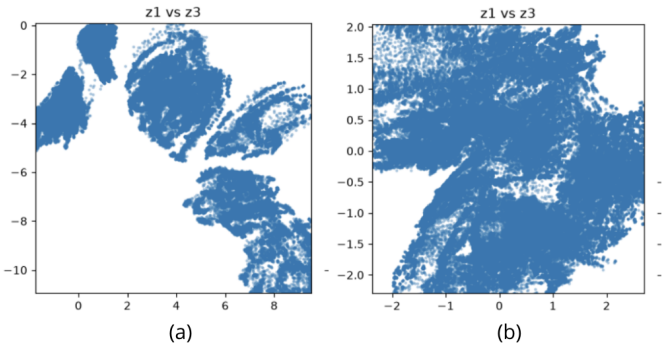


Fig. 2: Pairwise projections of the 4-dimensional latent space $(z_0, \ldots, z_3)$ for normal test samples. The AE latent space (a) shows elongated, curved manifolds, while the AAE latent space (b) is visibly more compact and isotropic, consistent with prior-matching toward a standard Gaussian.

Future work could explore testing different prior distributions, other than the standard Gaussian, to better match the physical properties of the robotic system. Also, using more advanced, non-linear models for score fusion could make the system even more reliable. Finally, testing this AAE + Fusion architecture in a real-time, edge-computing industrial setting would be an excellent next step to prove its usefulness for predictive maintenance.

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